

ASSIGNMENT – 5

Basic Statistics and Confidence Interval Analysis

1. Introduction

This assignment focuses on applying basic statistical techniques and confidence interval estimation to real-world datasets. The analysis includes descriptive statistics, data visualization, data preprocessing, and estimation of population parameters using confidence intervals.

2. Descriptive Statistics

Descriptive statistics summarize the central tendency and variability of numerical data. The following measures are computed: mean, median, mode, minimum, maximum, and standard deviation.

Metric	Description
Mean	Average value of the dataset
Median	Middle value of the dataset
Mode	Most frequently occurring value
Standard Deviation	Measure of data spread
Minimum & Maximum	Range of the data

3. Data Visualization

Histograms and boxplots are used to visualize the distribution and spread of numerical variables. Bar charts are used to represent categorical variables. These visual tools help identify outliers, skewness, and overall data patterns.

4. Data Preprocessing

Numerical variables are standardized so that they have a mean of zero and a standard deviation of one. Categorical variables are converted into dummy variables using one-hot encoding. This preprocessing step prepares the dataset for further statistical analysis and modeling.

5. Confidence Interval Estimation

Confidence intervals provide a range of values within which the true population mean is expected to lie with a specified level of confidence. In this assignment, a 99% confidence level is used.

Using Sample Standard Deviation (t-distribution)

When the population standard deviation is unknown and the sample size is small, the t-distribution is used to calculate the confidence interval.

Using Known Population Standard Deviation (Z-distribution)

When the population standard deviation is known, the Z-distribution is applied to estimate the confidence interval.

6. Conclusion

This assignment successfully applied descriptive statistics, data visualization, data preprocessing, and confidence interval estimation techniques. The analysis provides meaningful insights into the dataset and ensures statistically sound conclusions. All computations and visualizations were implemented using Python in Assignment-5.ipynb.